

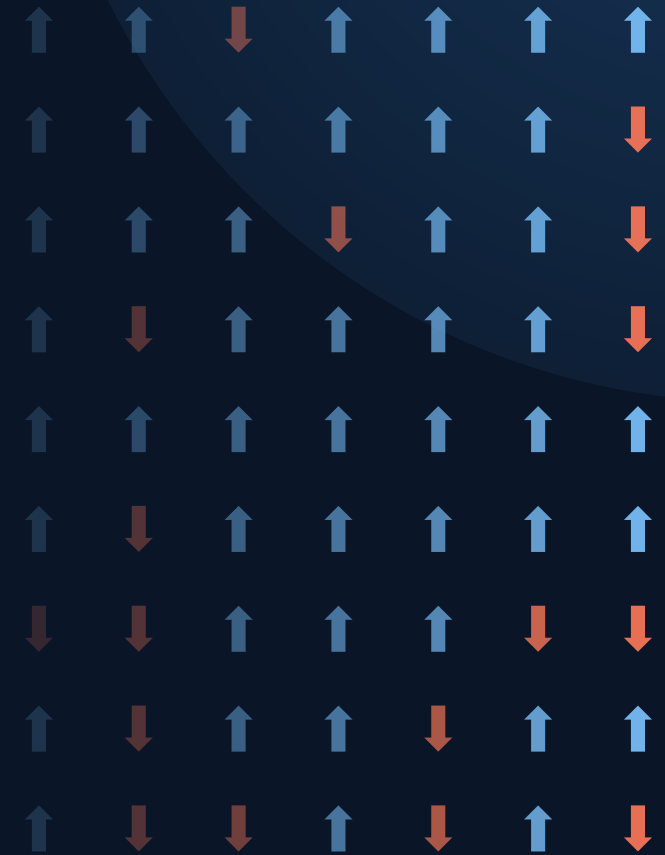
Measuring the External Field

Can we measure how hard a news article pushes on what people think?

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each arrow is one person's opinion



Physics treats media as a push, h , on opinion. Nobody measures h

$$m = \tanh(\beta J m + h)$$

The mean-field opinion model: the same maths as magnets lining up.

m average opinion of the group: -1 everyone against, +1 everyone for.

J coupling: how strongly each person is pulled toward the people around them.

β how firmly people follow that pull instead of acting at random (one over the noise).

h external field: the push from media, the same on everyone.



Arrows are people. Gold arrow is media, pushing everyone the same way.

Every study we found picks h by hand. **Our question: can h be measured from the content itself?**



What the proposal assumed, and what testing the tools showed

THE PROPOSAL ASSUMED

We would measure the amygdala, the brain's alarm centre.

The score would be a ratio of the two areas.

Outrage articles would stand out the most.

We would measure α (alpha), the strength of media's push.

The score would detect manipulative articles.

WHAT WE FOUND

The model only predicts the brain's outer surface.

Averages sit near zero, so the ratio broke for 69 of 400 articles.

Outrage raised both areas by the same amount.

The data could not pin it down.

Simple word counting did better.

WHAT WE DID

Used emotion-linked and reasoning-linked surface areas.

Used a simple difference: emotion minus reasoning.

Reported it; fear is what stands out.

Worked out the minimum α instead.

Presented it as a measurement, not a detector.

We also corrected two coefficients in the proposal's physics equation. All changes are in the approved amendment.



The tools we used, and why each one

TRIBE v2

Meta research, 2025

An AI model that predicts brain activity from text, sound and video. It learned from brain scans of adult volunteers watching TV and films.

Text-to-speech

reads each article aloud

The model was built on people listening, not reading, so every article is spoken first.

Word timing tool

forced alignment

Marks the moment each word is spoken, so the words and the sound line up for the model.

Brain area map

Glasser et al., 2016

A standard map that names areas of the brain's surface. We used it to pick emotion and reasoning areas.

400 public articles

four published collections

Anyone can download the same articles and check every number.

Cloud GPU + Python

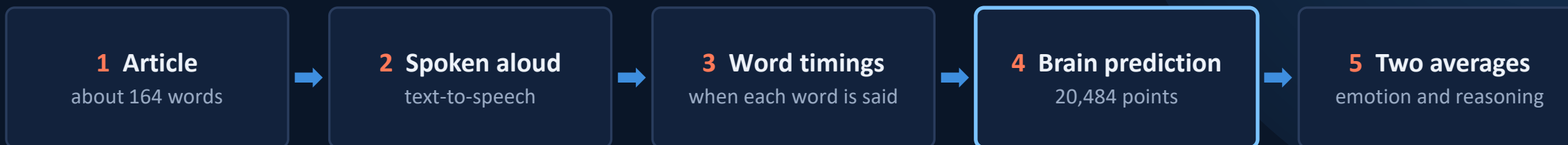
free Kaggle Tesla P100

About 70 seconds per article. All code and results are public on GitHub.

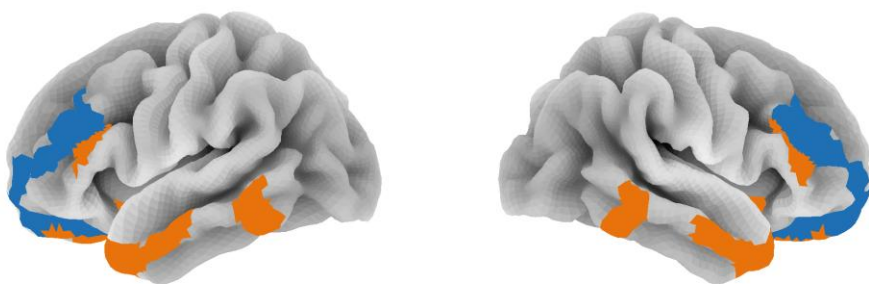
No person was scanned. Every brain value in this work is a prediction from the model.



How one article becomes one number



ROI definition, not a measurement: affective-salience (1030 vertices, orange) and deliberative-control (851, blue)
left right



Orange: 1,030 emotion-linked points. Blue: 851 reasoning-linked points.

THE SCORE, X

$$X = \bar{A}_{\text{emotion}} - \bar{A}_{\text{reasoning}}$$

$\bar{A}$ = average predicted activity over that set of points.

Above 0: emotion areas predicted more active.

Below 0: reasoning areas predicted more active.

20,484 = points on a standard model of the brain's surface, where the prediction is made.



400 articles in four groups, matched in length

100

Fear-driven

fake news collection (ISOT)

100

Outrage

hyperpartisan news (SemEval-2019)

100

Clickbait

clickbait headlines (Webis-17)

100

Neutral

medical abstracts + real news

≈ 164

words per article, on average

What it means: All four groups are within 5 words of each other.

Why this value: So a longer article can't score differently just by being longer.

0.027

smallest effect the study could see

What it means: We sized the study before scanning a single article.

Why this value: 400 articles is enough to catch a difference as small as 2.7%.

× 2

every article scanned twice

What it means: Two separate runs on the GPU.

Why this value: Lets us measure how repeatable the score is.



Result 1: the four groups really do differ

HOW MUCH THE GROUPS DIFFER (η^2)

0.107

$$\eta^2 = \frac{\text{variation between groups}}{\text{total variation}}$$



The second run repeats the result. Both are far above what we could detect.

$\approx 11\%$

$\eta^2 = 0.107$: share of the differences explained by group

What it means: Knowing an article's group explains about 11% of why its score differs from the others.

Why this value: The groups overlap a lot; most variation is from one article to the next.

1 in a billion

$p = 1.0 \times 10^{-9}$: the chance this is luck

What it means: If the groups were truly the same, a pattern this strong would appear by chance about once in a billion tries.

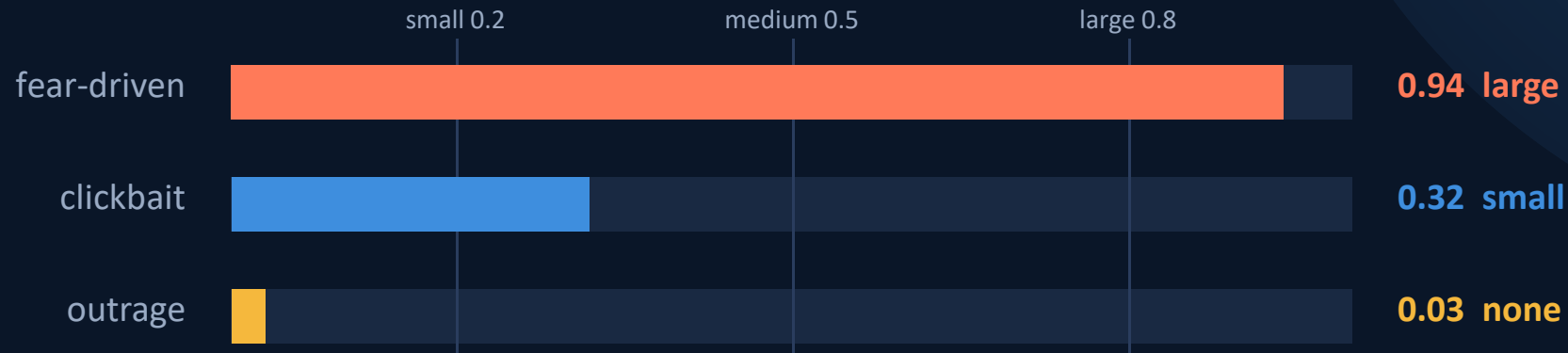
Why this value: 400 articles and a consistent difference between groups.



THE RESULTS

Fear drives the difference; outrage barely moves the score

DIFFERENCE FROM NEUTRAL ARTICLES (COHEN'S d)



$$d = \frac{\bar{X}_{\text{group}} - \bar{X}_{\text{neutral}}}{s}$$

$\bar{X}$ = average score of a group. s = normal spread of scores.

What it means: 0.2 is small, 0.5 medium, 0.8 large. Fear articles sit almost one full spread away from neutral.

WHY OUTRAGE STAYS FLAT

Outrage raised emotion areas by **0.51** and reasoning areas by **0.49**. The score is the gap between them, so it barely moves. Fear raised emotion areas only (0.61 vs -0.10).



The score repeats well for groups, not for single articles

0.87

Agreement between the two runs (ICC = 0.8725)

What it means: 1 means identical, 0 means no agreement. 0.87 is high.

Why this value: The speech voice and the computer's arithmetic vary a little between runs.

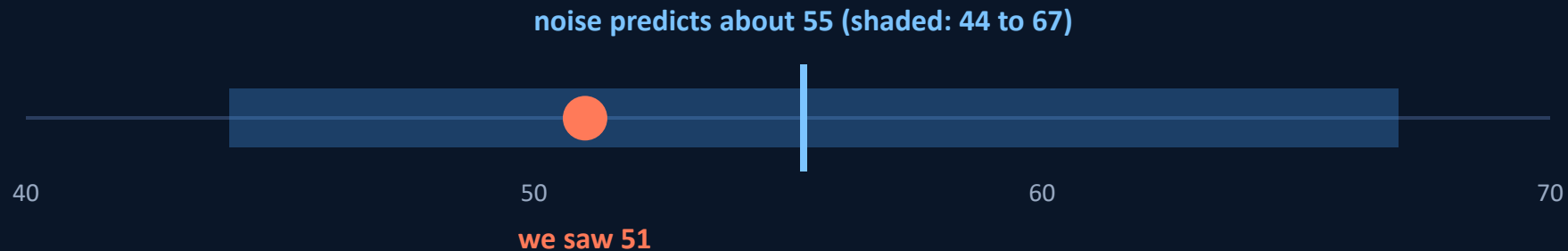
51 of 400

Articles whose score flipped sign (12.8%)

What it means: Noise alone predicts about 55. The ones that flipped all sit near zero.

Why this value: A score near zero tips either way with a tiny wobble, so we never judge one article.

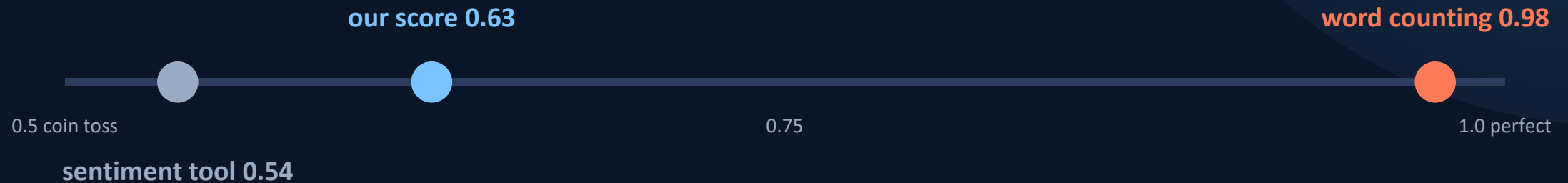
FLIPPED ARTICLES: WHAT WE SAW AGAINST WHAT NOISE ALONE PREDICTS





Honest check: simple word counting sorts articles better than our score

AUC = how well a score sorts manipulative from neutral articles. 0.5 is a coin toss, 1.0 is perfect.



WHY WORD COUNTING WINS

Each group came from a different source. From the words alone, a computer guesses the source 66% of the time, against 25% by chance. So it is partly spotting the source, not the style.

WHAT THIS MEANS FOR US

We cannot say the group differences come from writing style alone. Our score is offered as a measurement for the physics model, not as a manipulation detector.



Does the brain model match real brains? Weakly, but yes

A company audit claimed this model is the opposite of real brains. We tested it on public brain scans of people watching the TV show Friends.

0.152

Best possible score (noise ceiling)

What it means: How closely real people's brains agree with each other watching the same episode.

Why this value: Brains differ and scanners are noisy, so even a perfect model can't score higher.

0.028

Model vs real brains (r), 2-minute clip

What it means: The model's predictions rise and fall with real brain activity, weakly.

Why this value: The best possible on that same clip is 0.096, so the model gets under a third of the way.

$p = 0.048$

Chance the match is luck

What it means: Just under the usual 5% cut-off.

Why this value: Two minutes of video is short. A longer run is needed to settle it.

Along the way we found and fixed three hidden bugs: the wrong episode loaded, a table read sideways, and two clocks out of step.



The physics result: the minimum push media would need

$$h = \alpha X \Rightarrow \alpha \geq \frac{h_c(\beta J)}{\Delta X}$$

h the push from media on everyone (the field).

X our score for an article, emotion minus reasoning.

α strength: how hard one unit of score pushes. Not measured, so no value is quoted.

$h_c(\beta J)$ critical field: the smallest push that flips the majority. It grows with copying strength.

βJ copying strength. Above 1 a group settles on a majority by itself; 2 means strongly connected.

ΔX spread of our scores: from -0.070 to +0.054, so 0.124.

$$\alpha \geq 4.29$$

Minimum strength when copying is strong ($\beta J = 2$)

What it means: Any study saying content like this flips opinion must use an α of at least 4.29, or the maths says it can't happen.

Why this value: Our scores cover a narrow range (0.124), so each unit of score must push hard.

WHAT IT DOES NOT SAY

It does not claim that media flips opinion. It is a bar any such claim has to clear. The more strongly people copy each other, the higher the bar.



Can it analyse any content, even what children watch?



WHAT IT CAN DO

- Take video and sound as well as text, so a cartoon or an advert can be scored.
- Compare groups of content, like two channels or two kinds of programme.
- Score the content, never the viewer. Nobody is scanned.



WHAT IT CANNOT DO YET

- Say how a child reacts. The model learned from adult brains, and children's brains are still developing.
- Speak for video: only news text read aloud was tested here.
- Rate one show: about 1 in 8 items flip between runs.



WHAT IT WOULD TAKE

- Brain scans from children, collected with ethics approval and consent.
- A test set of children's videos.
- Content where source and type are mixed.
- Safeguards so it is never used to target children.

Short answer: technically possible, but not yet valid for children.



What this work cannot claim yet

LIMIT	WHY IT MATTERS	NEXT STEP
Predictions, not brain scans	Nobody was scanned; the model could be wrong.	The real-brain check (Paper 3).
Group is mixed up with source	Each group came from one collection.	A new set where every source has every type.
Brain surface only	No amygdala or other deep areas.	A model that predicts deep areas.
Strength of the push (α) unknown	The data can't pin it down.	Use the minimum rule instead.
The real-brain check is short	Only 2 minutes of video so far.	Longer clips, run one at a time.



Three papers, and what I need from you today

PAPER 1 · Physica A

The minimum push

How strong media would have to be to flip a majority. Pure physics; needs no brain data.

12 pages, ready

PAPER 2 · Physica A

The measuring tool

The process, the 400 articles, the results, and the word-counting comparison.

16 pages, ready

PAPER 3 · Imaging Neuroscience

Does the model match real brains?

The best-possible score, the three bugs, and the first check.

8 pages, one longer run to go

TODAY I AM ASKING FOR

- 1 Any corrections you need before the thesis is final.
- 2 Your signature on the declaration page (page iii).
- 3 Clearance to deposit with the library (bound copy by Thursday 24 September).
- 4 Your view on co-authorship of the three papers.